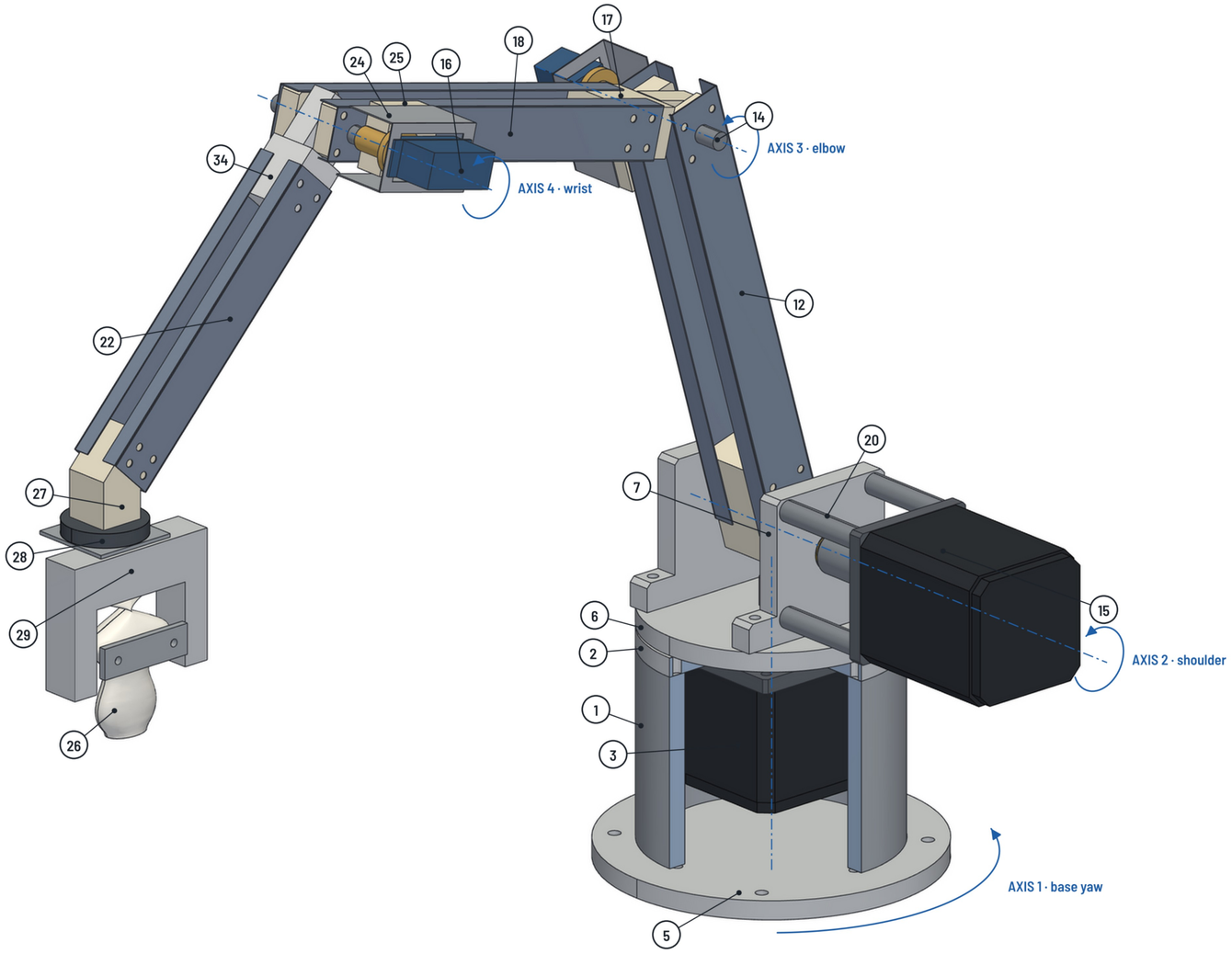
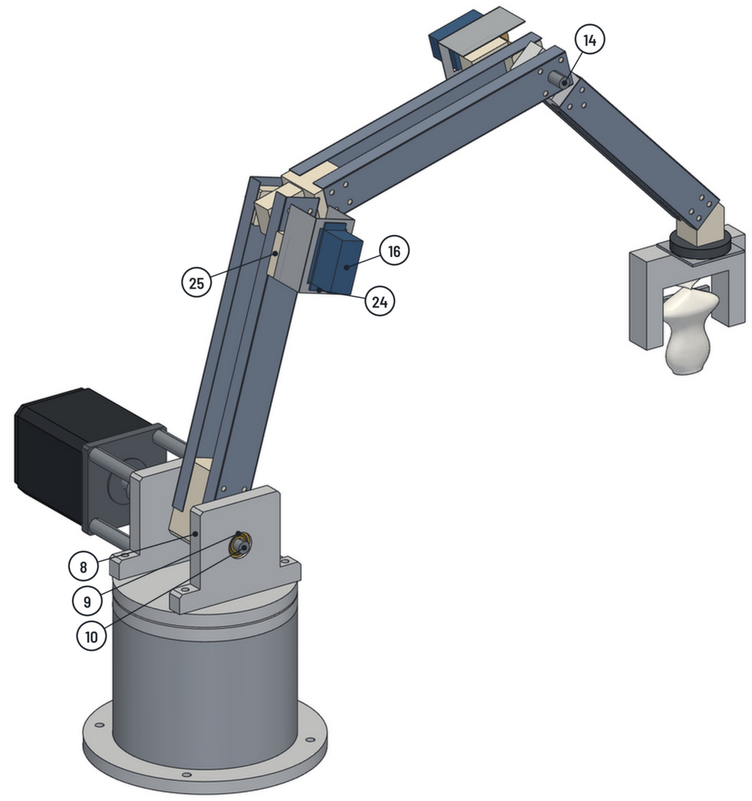
Robotic ultrasound arm, as designed

Four driven axes: base yaw plus shoulder, elbow and wrist pitch. Redrawn to scale from the final assembly drawing of 28-08-2023.

Robotic US System · BTP
Isometric views, not to scale
Blue dash-dot lines are the joint axes



Front-right view. The base housing is cut away to show the Axis-1 motor (3) hanging under the clamping plate (2).



Rear-left view: the shoulder bearing (9) and shaft (10), and the elbow servo (16) in its housing (24, 25).

Axes and drives, as listed

AXIS	MOTION	ACTUATOR	TRANSMISSION	BEARINGS
1	Base yaw	Stepper, 20 N·m (NEMA-42)	Direct, via turntable (6)	SKF 51209 thrust (4)
2	Shoulder pitch	Stepper, 12 N·m (NEMA-42)	Direct, via coupler 2 (19)	2 × SKF 16002 (9)
3	Elbow pitch	Servo DS5180, ≈80 kgf·cm	Direct, via coupler 3 (23)	2 × SKF 16002 (30)
4	Wrist pitch	Servo DS5180, ≈80 kgf·cm	Direct, via coupler 3 (23)	2 × SKF 16002 (9)

Parts

Item numbers as in the final-assembly parts list. 31, 32, 33 and 35 are the mirror-hand twins of 13, 18, 21 and 22.

BASE · AXIS 1

- 5 Base plate · Al Ø280 × 16
- 1 Base motor housing · Al Ø212 × 160
- 2 Axis-1 clamping plate · Al Ø212 × 16
- 3 Axis-1 motor · stepper, 20 N·m
- 4 Thrust bearing · SKF 51209
- 6 Axis-1 coupler / turntable · Al

SHOULDER · AXIS 2

- 7 Pillow block, motor side · Al
- 8 Pillow block, other side · Al
- 9 · 30 SKF 16002 bearings, 2 per joint
- 10 Axis-2 shaft · MS Ø15 × 170
- 11 Connecting block · nylon
- 15 Axis-2 motor · stepper, 12 N·m
- 19 Coupler 2 · MS Ø36 × 56
- 20 Motor spacers · MS Ø18, 4 off

UPPER ARM · AXIS 3

- 12 Link 1 · MS channel 49 × 20, 2 off
- 13 Bearing blocks 2 · nylon
- 14 Axis-3 shafts · MS Ø15
- 17 Axis-3 link

SERVO DRIVE · AXES 3 AND 4

- 16 Servo · DSServo DS5180
- 23 Coupler 3 · servo to shaft
- 24 Servo housing · 1 mm MS bracket
- 25 Servo housing block · nylon

FOREARM AND WRIST · AXIS 4

- 18 Link 2 · MS channel, 2 off
- 21 Bearing blocks 3 · nylon
- 34 Axis-5 link (wrist output)
- 22 Link 3 · MS channel, 2 off

END EFFECTOR

- 27 Probe holder 2 · nylon, 45° bend
- 28 Sensor bracket + 6-axis force sensor
- 29 Probe and bracket assembly · U-frame
- 26 "ID_1" imported part, drawn as the probe

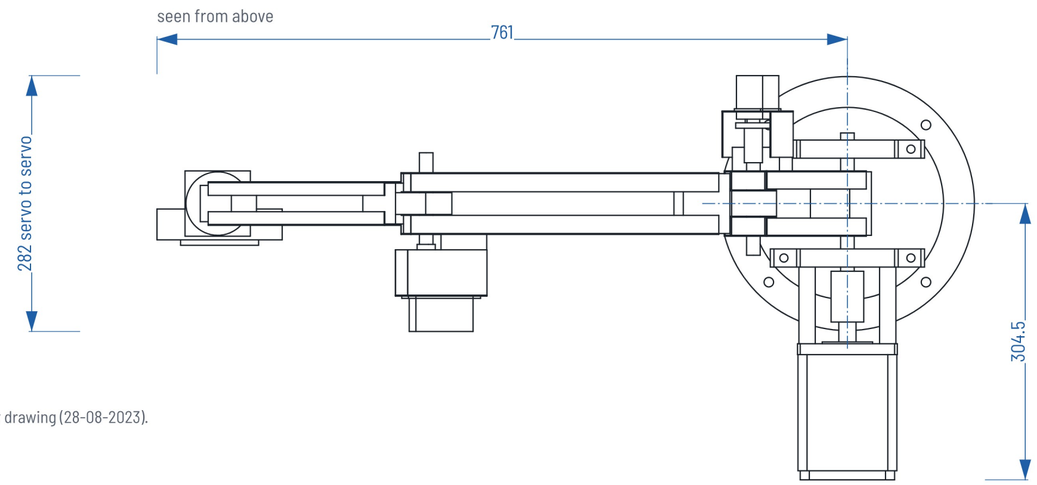
Plan, end view and side elevation

Hidden-line views of the same model, to scale, with the main dimensions. Pose as on the final assembly drawing.

Robotic US System · BTP
Third-angle projection
Dimensions in mm

PLAN

seen from above

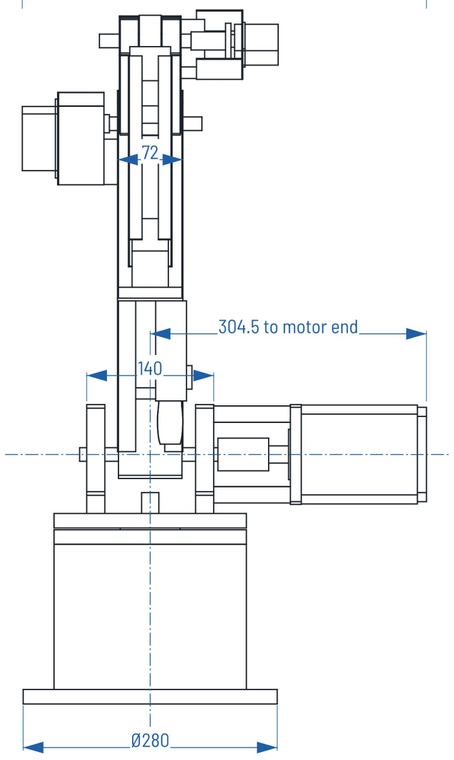


Dimensions in mm. Joint centres measured from the final assembly drawing (28-08-2023).

0 100 200 mm

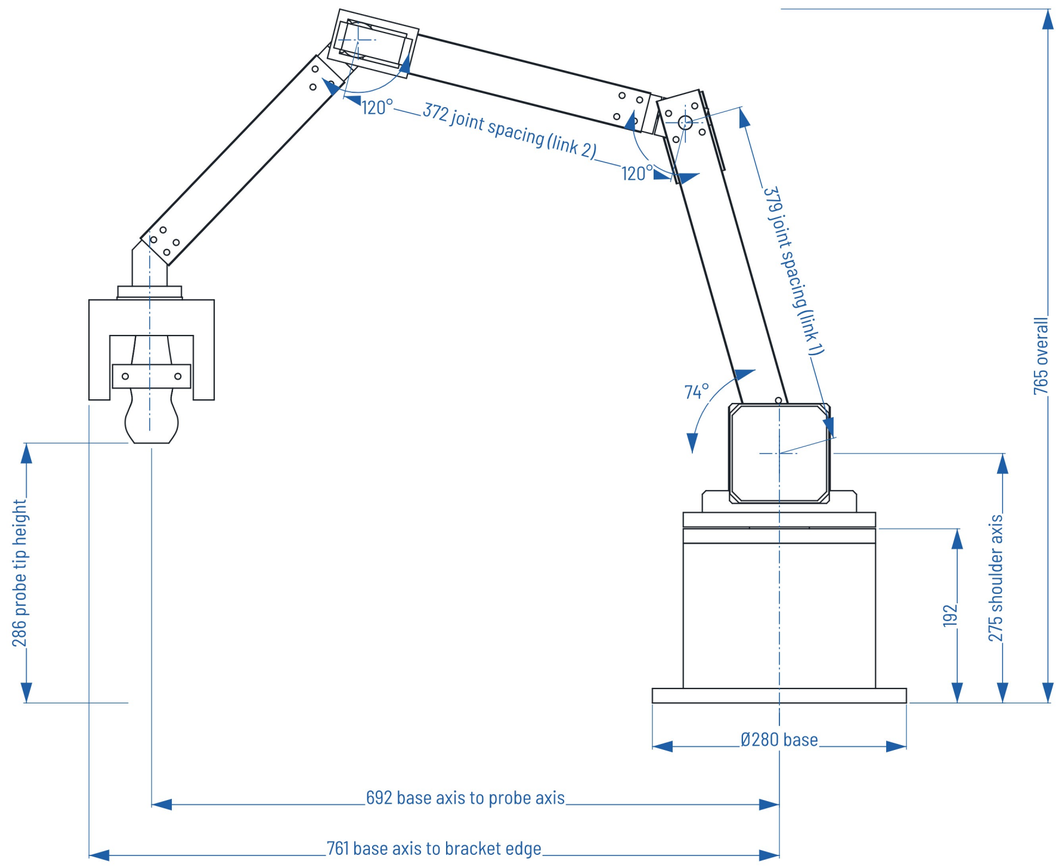
END VIEW

seen from the probe side



SIDE ELEVATION

seen from the Axis-2 motor side · drawn pose of the final assembly



Key dimensions

Shoulder axis above base underside	275
Joint spacing, shoulder to elbow (link 1)	379
Joint spacing, elbow to wrist (link 2)	372
Wrist axis to probe axis, horizontal	228
Base axis to probe axis	692
Probe tip height, drawn pose	286
Overall height	765
Overall length, plan	901
Overall width	445.5
Base plate	Ø280 × 16
Base motor housing	Ø212 × 160

Drawn pose

Link 1 at 74° above horizontal, 120° between the links at the elbow and at the wrist, probe pointing down. This is the pose of the final assembly drawing, not a limit of travel.

Note on link lengths

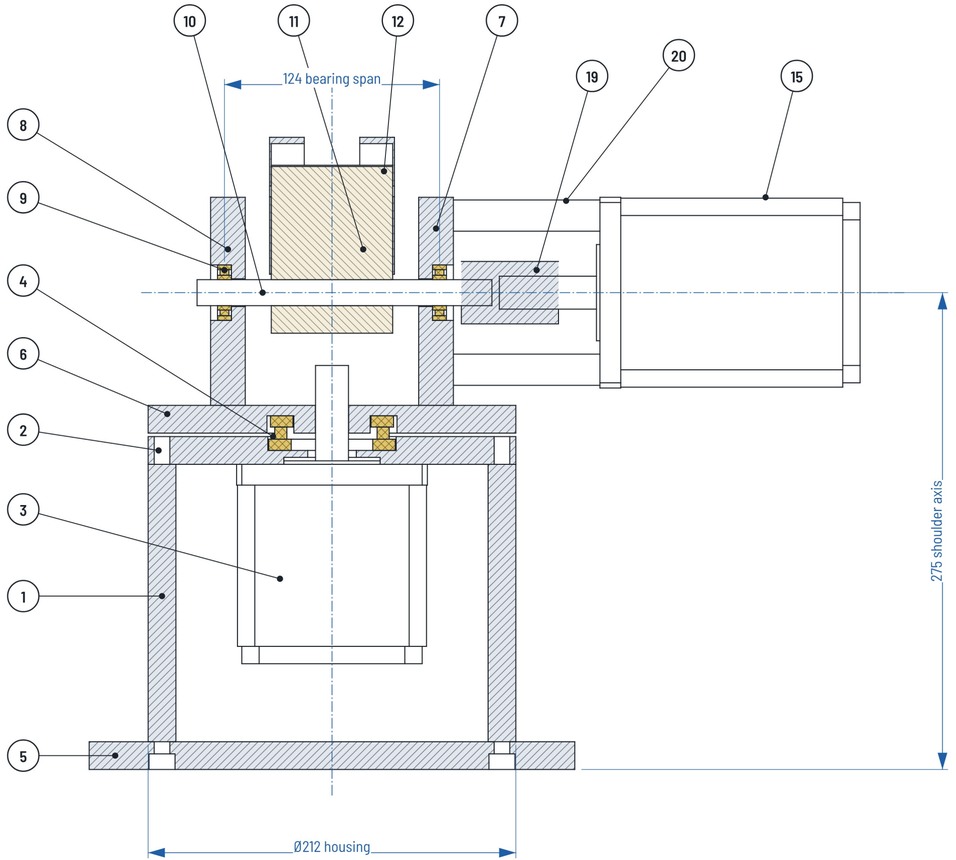
The 400, 350 and 280 mm in the parts list are the lengths of the bent sheet-metal channels, not joint-to-joint distances. Measured between joint centres on the final assembly drawing, the spacings are 379 mm (shoulder to elbow) and 372 mm (elbow to wrist).

The design-check report (revision B) now uses these measured spacings. The joint torques moved by 5 % or less, and none of its conclusions changed.

How the joints are built

Sections through the base and shoulder and through the elbow, and a side view of the end effector.

Each view at its own scale
Dimensions in mm
Item numbers as on sheet 1

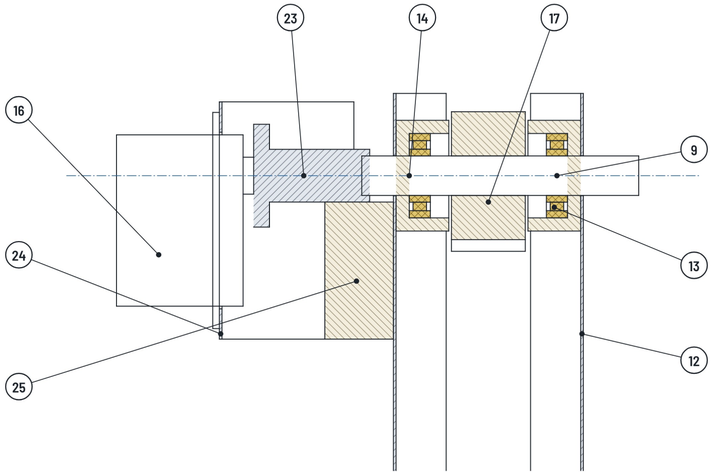


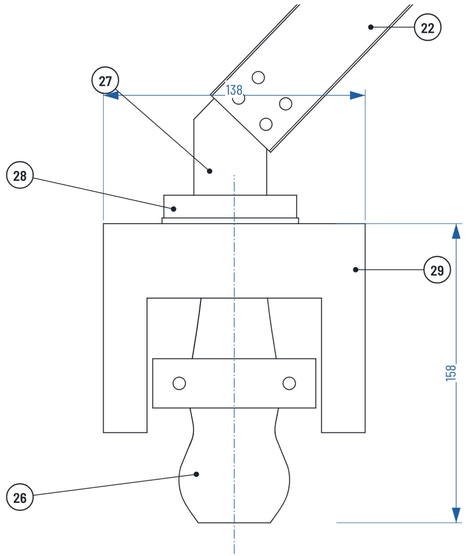
SECTION A–A · base and shoulder (Axes 1 and 2)

cut through the base axis and the shoulder axis, seen from the probe side

SECTION B–B · elbow (Axis 3)

cut along link 1 through the elbow axis; the wrist (Axis 4) is built the same way, servo on the other side





DETAIL C · end effector

side view; probe held in a U-frame under the force sensor

A–A · Base (Axis 1)

The Axis-1 motor (3) hangs under the clamping plate (2) and its shaft carries the turntable (6). The thrust bearing (4) takes the weight but resists only about 4 N·m of tilting moment; the arm applies 18–28 N·m, so the rest loads the motor shaft. See sheet 4, A1.

B–B · Elbow (Axis 3)

The elbow shaft (14) runs in bearing blocks (13) held in link 1 (12). The servo (16) sits in its housing (24) on a nylon block (25) and drives the shaft through coupler 3 (23). The wrist is built the same way, mirrored.

A–A · Shoulder (Axis 2)

The shoulder shaft (10) runs in SKF 16002 bearings (9) in the pillow blocks (7, 8). The motor (15) stands on four spacers (20) and drives through the rigid coupler (19); the nylon block (11) keys link 1 to the shaft.

C · End effector

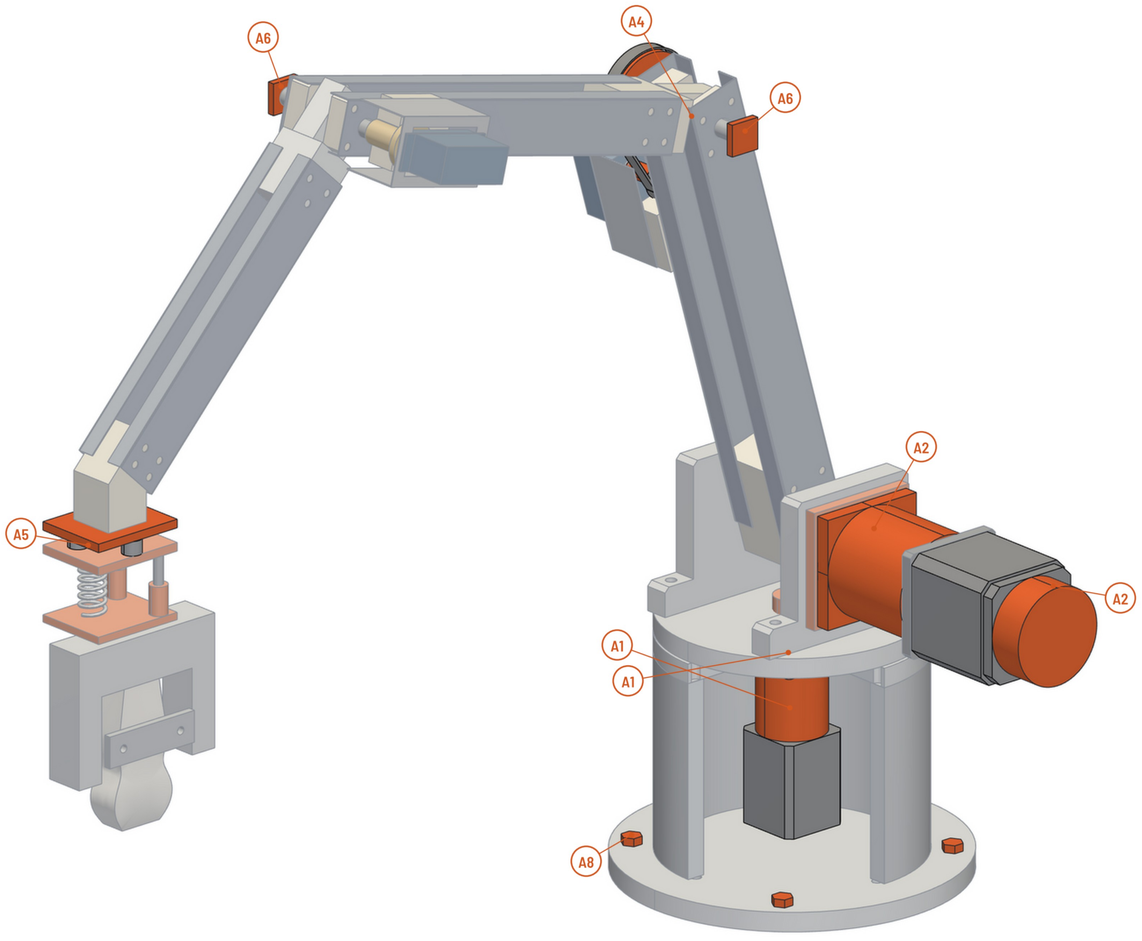
The 6-axis force sensor and its bracket (28) sit between probe holder 2 (27) and the U-frame (29) that clamps the probe (26). In the recommended build a load cell and spring replace the sensor (sheet 4, A5).

SECTION HATCHING — Aluminium and steel · Nylon · Bearings

Same arm, set up for fully autonomous scanning

Orange parts are new or changed; grey parts are as designed. This is the "Recommended" build in the cost estimate.

Robotic US System · BTP
Isometric view, not to scale
A1–A12 as in the design-check report



New or changed part As designed "Not drawn" items have no visible part in the view.

A1 Base bearing and drive DRAWN
Crossed-roller bearing (RU66 class) under the turntable; a geared NEMA-23 with a 50:1 high-precision planetary and a flexible coupling replaces the NEMA-42.
Why: the thrust bearing resists only ≈4 N·m of tilting moment; the arm applies 19–28 N·m.

A2 Shoulder drive DRAWN
NEMA-34 stepper (8.5 N·m) with a 10:1 planetary gearbox and a 24 V power-off brake.
Why: the shoulder needs ≈13.6 N·m in the drawn pose and ≈25 N·m stretched out; the 12 N·m motor gives ≈8.4 N·m usable.

A3 Shoulder shaft and hub NOT DRAWN
EN19/EN24 shaft with k5/H7 bearing fits; a clamp hub in place of the keyed nylon block (11).
Why: the geared drive puts more torque through the shaft; a keyed nylon hub wears and adds play.

A4 Elbow reduction DRAWN
3:1 HTD timing-belt stage between the DS5180 servo and the elbow shaft; aluminium bearing blocks.
Why: the elbow needs ≈9.8–10.3 N·m in the scanning zone; the servo alone gives ≈5.1 N·m usable.

A5 Force and tilt sensing DRAWN
Three load cells under the probe holder, over a spring stage (0.5–1 N/mm, two guide rods), replace the 6-axis force/torque sensor.
Why: they measure the probe force and the two tilt moments used to keep the probe normal; far cheaper and lighter, about 1 N resolution.

A6 Joint encoders with position loops DRAWN ON AXES 2-4
AS5048-class absolute encoders on all four joints, read after the gearboxes (the base one through a 1:1 belt), with position loops closed on them; homing switches on Axes 1 and 2.
Why: pressing the probe with 7–20 N cancels the arm's weight at the joints, so gear play and servo dead band show up exactly while scanning.

A7 Servo signal NOT DRAWN
Servo PWM at 330 Hz or more, or timer steps of 1 µs or less.
Why: a 50 Hz PCA9685 moves the servos in 0.44° steps; this brings them down to about 0.07°.

A8 Base fixing DRAWN
Bolt the base plate down with 4 × M10 on the Ø245 PCD.
Why: standing free, the tipping safety factor at full reach is only about 1.5.

A9 Safety NOT DRAWN
Brakes, an E-stop that holds position and then brakes, a 25 N force limit, software joint limits and a watchdog.
Why: as designed, the arm drops when the power is cut, and nobody is at the controls during a scan.

A10 Probe roll, optional NOT DRAWN
A small servo (20–35 kgf·cm) turns the probe about its own axis as a fifth axis.
Why: switches between transverse and longitudinal views without moving the arm.

A11 Finding the body NOT DRAWN
A depth camera over the scan area, calibrated to the robot with a printed board — or, for a phantom of known shape, touch it at about 25 points.
Why: an autonomous robot has to find the surface and its normals itself before it can plan the sweeps.

A12 Ultrasound image to the PC NOT DRAWN
USB video capture of the scanner's output.
Why: every frame is logged with the probe pose, and lost contact shows up as a dark image the planner can react to.

What the changes buy

Over the scanning zone (r 400–750 mm, z 100–300 mm), no point is within the usable torque of the motors as designed; with Option A, every point is. Estimated probe-tip error: ≈13 mm as designed, ≈5 mm with Option A on its own (gearbox play included), ≈1.7 mm with position loops on the joint encoders.

Placed beside the phantom, the arm can hold the probe within 15° of the surface normal over about 71 % of its upper surface. It cannot tilt the probe sideways, so the steep flanks need the robot moved or the phantom turned.